

Working scientifically questions KS4

Q1

0 3

Amylase is an enzyme that digests starch.

A student investigated the effect of pH on the activity of amylase.

This is the method used.

1. Mix amylase solution and starch suspension in a boiling tube.
2. Put the boiling tube into a water bath at 25 °C.
3. Remove a drop of the mixture every 30 seconds and test it for the presence of starch.
4. Repeat the investigation at different pH values.

Table 1 shows the students' results.

Table 1

pH	Time when no starch was detected in minutes
5.0	7.0
5.5	4.5
6.0	3.0
6.5	2.0
7.0	1.5
7.5	1.5
8.0	2.0

What conclusion can be made from the table.

- A. The optimum pH for amylase activity is pH 7.
- B. The optimum pH for amylase activity is pH 7.25
- C. All pH between 7 and 7.5 give the same amylase activity.
- D. More readings at smaller intervals are needed to identify the optimum pH

Answer D

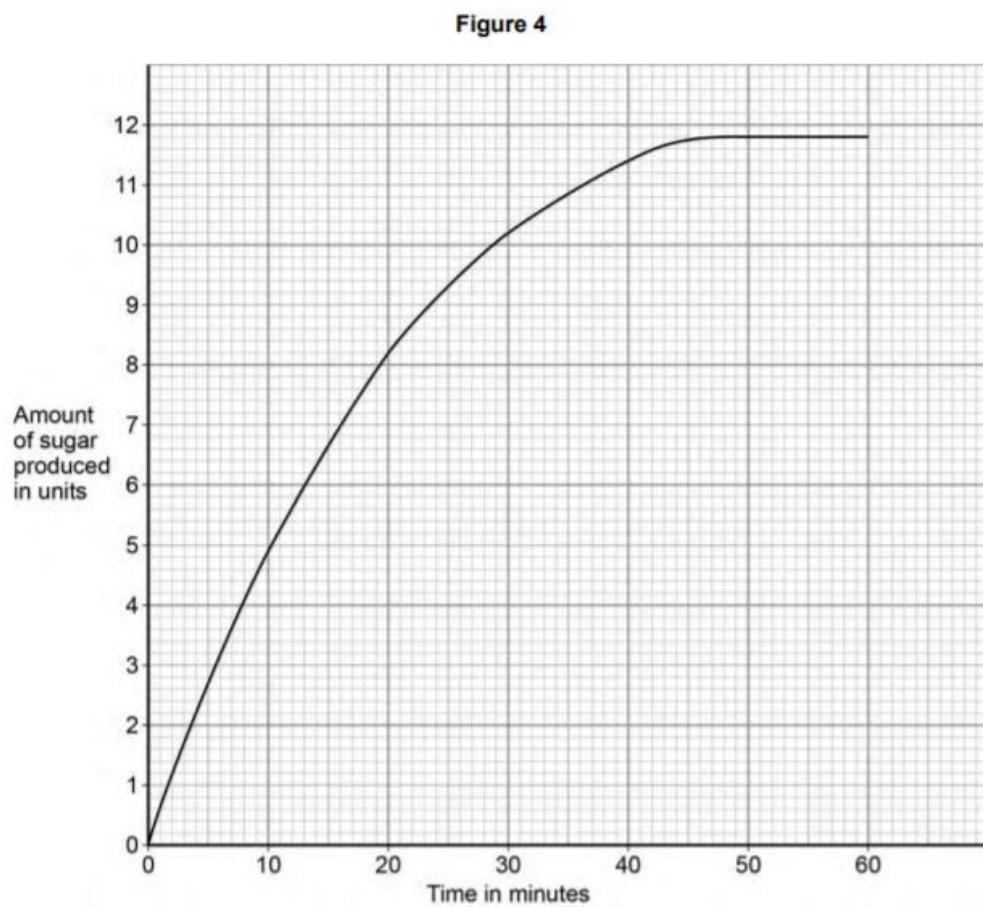
Question 2

The student did another investigation.

This is the method used.

1. Put amylase solution and starch suspension into a boiling tube.
2. Make the pH 7.25.
3. Put the boiling tube into a water bath at 25 °C.
4. Measure the amount of sugar produced every 30 seconds.

The results are shown in **Figure 4**.



What is the mean rate of sugar produced in the first 5 minutes?

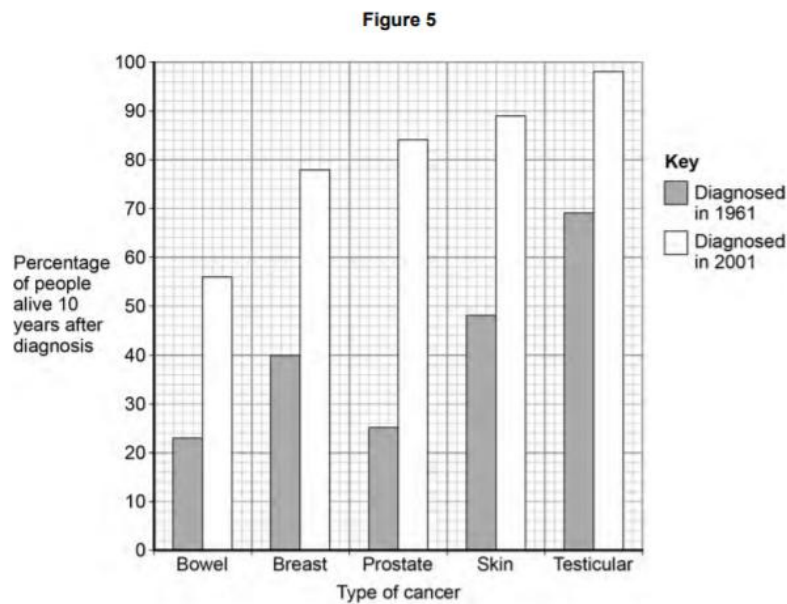
- A 2.7
- B 1.81
- C 0.54
- D 0.197

Answer C

Question 3

Some people who are alive 10 years after diagnosis are considered to be cured.

Figure 5 shows data for people diagnosed with cancer in 1961 and 2001.



SPECIMEN MATERIAL

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0 4 . 2 Look at the data in **Figure 5** for skin cancer.

Calculate the percentage increase in the survival rate of people diagnosed with skin cancer in 1961 compared to 2001.

Give your answer to **three** significant figures.

- A: 85.6%
- B 40.0%
- C 48.0%
- D 46.1%

Answer A

Question 4

Two students investigated the effect of caffeine concentration on reaction time.

This is the method used.

1. Student **A** drinks a cup of coffee.
2. Student **B** holds a ruler above Student **A**'s hand.
3. Student **B** drops the ruler.
4. Student **A** catches the ruler as quickly as she can.
5. The distance the ruler falls is recorded.

The method for the experiment can be improved. Which of the following would not improve the validity of the results collected?

- A. Repeating the results with decaffeinated coffee
- B. A blind trial with a variety of types of coffee
- C. Using a data logger to measure the distance.
- D. Changing the concentration of coffee

Answer C

Question 5

Two alleles control the body colour of carp:

- brown (**B**)
- blue (**b**).

The brown allele is dominant to the blue allele.

The genetic cross from breeding two carp is shown in **Figure 8**.

Figure 8

	B	b
b	Bb	
b		

What is the probability that the offspring from this genetic cross will be brown?

0

0.25

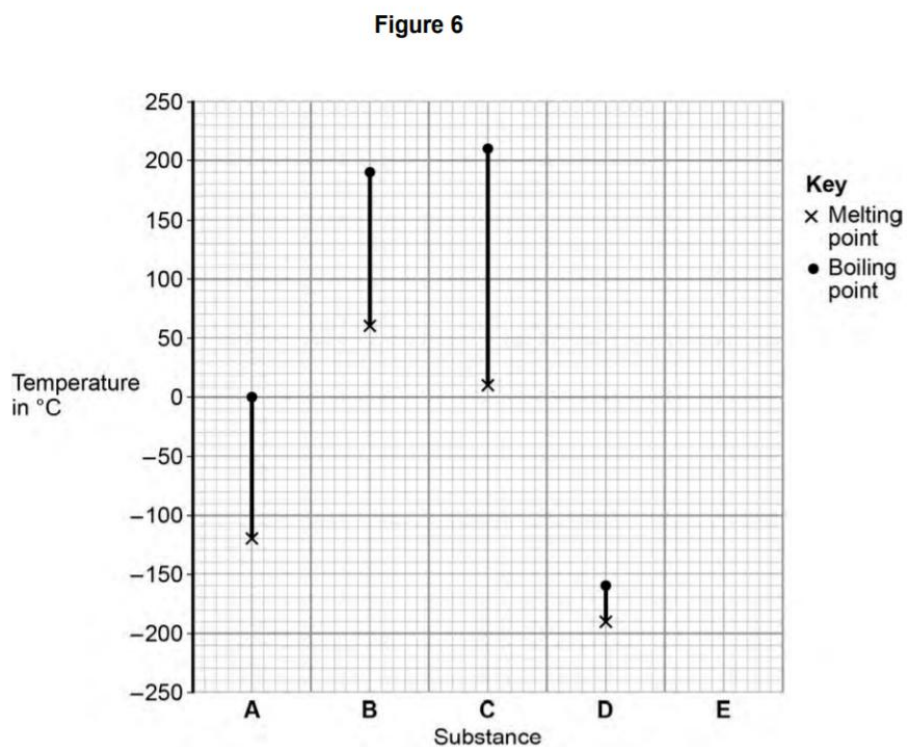
0.5

1.0

Answer C

Question 6

Figure 6 shows the melting points and the boiling points of four substances, **A, B, C** and **D**.



Which **two** substances are gases at 50 °C?

Tick **one** box.

A and B ☐

B and C ☐

C and D ☐

A and D ☐

Answer D

Question 7

0 3 . 6

A sample of drinking water contains 1.5 mg of fluoride per dm^3 of water.
A person drinks 1 dm^3 of this water.

The recommended daily amount of fluoride is 4.0 mg.

Which calculation gives the percentage of the recommended daily amount of fluoride in 1 dm^3 of this water?

[1 mark]

Tick **one** box.

$$\frac{1.5 \times 100}{4.0}$$

☐

$$\frac{1.5 \times 4.0}{100}$$

☐

$$\frac{4.0 \times 100}{1.5}$$

☐

$$\frac{100}{1.5 \times 4.0}$$

☐

Answer B

Question 8

A newtonmeter measures the weight of objects.

Look at **Figure 3**.

Figure 3



A student carried out a practical to investigate the extension of a spring.

What could be done to improve the accuracy in this investigation?

Tick **two** boxes.

Use a pointer from the spring to measure the length.

☐

Use a stronger spring in the practical.

☐

Use a new spring between each reading.

☐

Make sure the spring is stationary before measuring length.

☐

Use a longer rule when measuring length.

☐

Answer A and D

Question 9

The equation $\text{weight} = \text{mass} \times \text{gravitational field strength}$ shows the relationship between weight and mass.

The unit of mass is Newtons, the unit of mass is Kg. What is the unit of gravitational field strength

A N Kg^{-1}

B N Kg

C Kg N^{-1}

To get the mark you must show your working

Answer A $\text{weight} = \text{mass} \times \text{gravitational field}$

$\text{Gravitational field} = \text{weight} / \text{mass}$

Input units $G = \text{N/Kg}$ in scientific notation this is N Kg^{-1} .